



$$x_{cm} = 0$$

$$y_{cm} = \frac{y_{cm1} M_1 + y_{cm2} M_2}{M_1 + M_2} = \frac{\left(\frac{l}{2} \cos \frac{\pi}{4} + \frac{l}{2} \cos \frac{\pi}{4}\right) M}{2M} = \frac{l}{2} \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2} l$$

B, C e D sono allineati perché hanno la stessa y

$$\overline{OB} = \frac{\sqrt{2}}{2} l$$

$$\overline{OA} = d$$

$$\Omega = \frac{M g \overline{AB}}{I_z}$$

$$-M g \overline{AB} \sin \alpha = I_z \ddot{\alpha} \rightarrow \ddot{\alpha} + \frac{M g \overline{AB}}{I_z} \alpha = 0 \rightarrow \Omega = \frac{M g \overline{AB}}{I_z}$$

Dobbiamo trovare $\overline{AB}$ e I_z ($\overline{AB}$ e d)

$$\overline{AH} = d \sin \left(\frac{\pi}{4} \right)$$

$$\overline{BH} = \overline{OH} - \overline{OB} = d \cos \frac{\pi}{4} - \frac{l}{2} \cos \left(\frac{\pi}{4} \right) = \frac{\sqrt{2}}{2} (d - \frac{l}{2})$$

$$\overline{AB}^2 = \overline{AH}^2 + \overline{BH}^2 = \frac{d^2}{2} + \frac{1}{2} \left(d^2 + \frac{l^2}{4} - dl \right) = \frac{d^2}{2} + \frac{d^2}{2} + \frac{l^2}{8} - \frac{dl}{2} = d^2 + \frac{l^2}{8} - \frac{dl}{2}$$

Per calcolare I_z calcolo I_{cm} e poi uso H-S per calcolare $I_z = I_{cm} + M \overline{AB}^2$

$I_{cm} = 2 I_{cm}(\text{asta singola})$ perché è simmetrico

$$I_{cm} = 2 \left(I_{cm} + M \overline{CB}^2 \right) = 2 \left(\frac{M l^2}{12} + M \left(\frac{l}{2} \sin \frac{\pi}{4} \right)^2 \right) = 2 \left(\frac{M l^2}{12} + M \frac{l^2}{8} \right) = \frac{M l^2}{6} + \frac{M l^2}{4} = \frac{5}{12} M l^2$$

$$I_z = I_{cm} + 2 M \overline{AB}^2 = \frac{5}{12} M l^2 + 2 M \left(d^2 + \frac{l^2}{8} - \frac{dl}{2} \right) = \frac{5}{12} M l^2 + 2 M d^2 + \frac{l^2}{4} M - M l d = \frac{2}{3} M l^2 - M l d + 2 M d^2$$

$$\Omega = \frac{M g \sqrt{d^2 + \frac{l^2}{8} - \frac{dl}{2}}}{\left(\frac{2}{3} l^2 - l d + 2 d^2 \right) M}$$

Cerco il massimo di Ω facendo la derivata allora devo essere positiva, quindi il numeratore e il den. dovranno essere positivi:

$$\Delta_{num} = -\frac{1}{4} l^2 < 0 \quad \text{num} > 0 \quad \forall d$$

$$\Delta_{den} = l^2 - 5 l^2 < 0 \quad \text{den} > 0 \quad \forall d$$

$$\dot{\omega}^2(d) = 0 \rightarrow \frac{\frac{1}{2} (2d - \frac{1}{2} l) \frac{1}{\sqrt{\text{num}}} \cdot \left(\frac{2}{3} l^2 - l d + 2 d^2 \right) - \sqrt{\text{num}} (-l + 4d)}{(den)^2} = 0$$

$$\frac{\frac{1}{2} (2d - \frac{1}{2} l) \left(\frac{2}{3} l^2 - l d + 2 d^2 \right) - \left(d^2 + \frac{1}{8} l^2 - \frac{1}{2} l d \right) (-l + 4d)}{\sqrt{\text{num}}} = 0$$

raccolgo $\frac{1}{2}$ qui

$$\frac{1}{2} \cdot \frac{1}{2} (4d - l) \left(\frac{2}{3} l^2 - l d + 2 d^2 \right) - (4d - l) \left(d^2 + \frac{1}{8} l^2 - \frac{1}{2} l d \right) = 0$$

$$(4d - l) \left[\frac{1}{4} \left(\frac{2}{3} l^2 - \frac{l d + 2 d^2}{4} \right) - d^2 - \frac{1}{8} l^2 + \frac{1}{2} l d \right] = 0$$

$$4d - l = 0 \quad d = \frac{l}{4}$$

$$\frac{1}{6} l^2 - \frac{1}{6} l d + \frac{1}{2} d^2 - d^2 - \frac{1}{8} l^2 + \frac{1}{2} l d = 0$$

$$\frac{1}{24} l^2 + \frac{1}{6} l d - \frac{1}{2} d^2 = 0$$

$$-2 \left(\frac{1}{24} l^2 + \frac{1}{6} l d - \frac{1}{2} d^2 \right) = 0 \rightarrow -\frac{1}{12} l^2 - \frac{1}{3} l d + d^2 = 0 \rightarrow l \left(\frac{1}{6} \pm \frac{\sqrt{7}}{6\sqrt{3}} \right) = \frac{l}{6} \left(1 \pm \sqrt{\frac{7}{3}} \right) \approx \frac{l}{6} (1 \pm \sqrt{2}) \rightarrow \frac{l}{6} (1 + \sqrt{2})$$

Allora

$$d = \frac{l}{6} \text{ oppure } d = \frac{l}{6} \left(1 + \sqrt{\frac{7}{3}} \right)$$

metto dentro Ω questi valori e vedo quale dei due massimizza

$$R = \sqrt{\frac{I_z}{2M}} \quad \text{se} \quad \overline{AB}^2 = R^2 G \quad R \text{ raggio di girazione } \sqrt{\frac{I_{cm}}{2M}}$$

INUTILE $\rightarrow$ bastava fare $d = \sqrt{\frac{I_{cm}}{M}}$